

AKHILESH KUMAR MISHRA

Software Engineer / Full Stack Engineer

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PROFILE

Versatile Software Development Engineer with extensive experience building secure, scalable full-stack web applications and microservice architectures across multiple technology stacks (MERN, Java Full Stack, Python Full Stack). Expert in system design, API development, and database engineering. Proven success in re-architecting monolithic services into containerized microservices, building real-time telemetry pipelines via WebSockets, and optimizing transaction-heavy ledger platforms. Led development teams and streamlined DevOps workflows using Docker and CI/CD.

CORE TECHNOLOGY

Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

Frameworks: Spring Boot, Django, FastAPI, Node.js, Express.js, React.js, Next.js, Redux

Data & Messaging: MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, Mongoose, Hibernate, JPA

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, Postman, Linux, OpenCV, Pandas, Maven

EXPERIENCE

Software Development Engineer — CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Re-architected monolithic backend services to distributed microservices using **Spring Boot**, **FastAPI**, and **Node.js**, improving performance for **20 lakh+ users**.
- Architected transactional endpoints for AEPS, MATM, and UPI platforms, managing **5 lakh+ daily transactions** with an **~80% success rate**.
- Secured backend APIs utilizing **JWT Authentication** and **RBAC**, integrating device binding, rate limiting, and OTP security features.
- Designed and optimized databases using **MySQL**, **PostgreSQL**, and **MongoDB**, utilizing sharding, indexing, and query tuning.
- Established asynchronous task processing and event-driven communication using **Kafka**, **Celery**, and **Redis** cache.
- Integrated telemetry tracking and transaction monitoring dashboards using **React.js**, **Redux Toolkit**, and **WebSockets**.

Freelance Software Engineer — Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Designed and developed a real-time workforce telemetry and geofencing system using a **React** frontend and a **FastAPI / Node.js** backend.
- Built an interactive live tracking dashboard in **React** using Ant Design (antd) to display coordinates, speed, and alerts via WebSockets.
- Developed an asynchronous location ingestion pipeline utilizing **FastAPI**, **Celery**, and **Redis** to capture GPS coordinates.

SELECTED PROJECTS

MERN FinTech & E-commerce Platform

React.js / Node.js / Express.js / MongoDB / JWT / Redux

A collection of full-stack web applications featuring secure user authentication, interactive dashboards, and REST APIs built with React, Node, Express, and MongoDB.

Facial Recognition Attendance System

Python / OpenCV / Tkinter / MySQL / Pandas

Engineered an automated student attendance tracking application integrating Python face detection algorithms (OpenCV/LBPH) with MySQL database.

Himalayan Edges Website Development

React.js / Django / SQLite / AWS / PWA

Created an e-commerce platform using React.js and Django with PWA capabilities, hosted on AWS, increasing page speed by 30%.

Traveling Portal & Landing Page

HTML5 / CSS3 / JavaScript / Bootstrap

A fully responsive, modern front-end landing page for a travel agency designed with HTML, CSS, and Bootstrap layouts.

Persistent Task Management Dashboard

Python / Tkinter / JSON / MVC Patterns

Programmed a desktop utility for task planning, category filtering, and JSON data persistence, featuring clean MVC architecture patterns.

Real-Time Telemetry Geofencing Engine

React.js / FastAPI / Celery / Redis / PostgreSQL

Built a real-time workforce geolocation tracking engine with asynchronous GPS coordinate logs, map visualizations, and alerts via WebSockets.

EDUCATION

CERTIFICATIONS

GeeksforGeeks MERN Stack Certification
Coding Ninjas DBMS Certification
Coding Ninjas DSA Certification
IIT Kanpur Prutor Cyber Security Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.